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General Right Censoring and Its Impact on the Analysis of Survival Data

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Summary

This paper concerns general right censoring and some of the difficulties it creates in the analysis of survival data. A general formulation of censored-survival processes leads to the partition of all models into those based on noninformative and informative censoring. Nearly all statistical methods for censored data assume that censoring is noninformative. Topics considered within this class include: the relationships between three models for noninformative censoring, the use of likelihood methods for inferences about the distribution of survival time, the effects of censoring on the K-sample problem, and the effects of censoring on model testing. Also considered are several topics which relate to informative censoring models. These include: problems of nonidentifiability that can be encountered when attempting to assess a set of data for the type of censoring in effect, the consequences of falsely assuming that censoring is noninformative, and classes of informative censoring models.

1. Introduction

Censored survival data is fundamentally different from other types of data encountered in statistical problems in the sense that the response of interest, the time until some specified event, cannot always be fully observed. Instead, causes for censoring exist which can result in the interruption of observation before the event occurs. When this happens the recorded data do not provide direct information about the time until the event, and so models must be used to relate what one observes to what he wants.

In reliability studies, censoring is often planned for in order to obtain information sooner than is otherwise possible (*cf* Epstein and Sobel 1953). For example, rather than place r units on test and observe each until it fails, one can employ so-called Type I censoring designs in which $n(n > r)$ units are put on test and observation is terminated at the end of some specified period T^*. Those units which fail before T^* yield complete observations, whereas the rest give rise to right censored observations at T^*. This type of design can be used to obtain as much information as an uncensored experiment of size r and, by choosing n sufficiently large, it can be completed in much less time (*cf* Epstein 1954).

In clinical trials and other medical and biological studies, there are many more causes for and types of censoring than just the predetermined termination of an observation period. Although right-censored observations are the most common type, there can be left- and

Key Words: Censored observations; Survival time; Informative and noninformative censoring; Censoring patterns.

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doubly-censored observations as well. Censored observations also typically occur at numerous time points, hereafter referred to as *general* censoring, rather than at a single time point. The complexity of the situation is enhanced even further by the fact that there are often several different causes for censoring in effect in a single study.

My purpose in this paper is to discuss censoring and the problems it creates in the analysis of data for the characteristics of a survival-time distribution. An examination of the recent statistical literature will reveal that much research has and is being done on problems involving censored data. Although most of the results which have been reported pertain directly to the modeling and analysis of censored data, there are others which have led to new ideas about the use of likelihood functions for statistical inference and the relationships between likelihood and nonparametric methods (see, for example, Cox 1972, Kalbfleisch and Prentice 1973, Cox 1975, and Kalbfleisch 1978). Despite the number of useful results already obtained, a review of the literature will also indicate that many obviously important questions remain to be answered and that many ideas which are the bases for current practice need to be clarified and more rigorously developed.

In this paper, I will focus attention on issues which relate directly to censored data. Three models for noninformative censoring, the type assumed in nearly all statistical methods for censored survival data, are presented and compared in Section 3.1. The remainder of Section 3 addresses problems of estimation, comparisons of several samples, and tests of goodness-of-fit for the underlying survival time distribution in the presence of this type of censoring. Section 4 begins with a discussion of problems of nonidentifiability and observable indistinguishability that can be encountered in the evaluation of censored survival data for the type of censoring which is in effect. This is followed by an assessment of the consequences of incorrectly using a model based on noninformative censoring and the consideration of models for situations where censoring is related to survival time.

The principal goal of this paper is to provide an outline of some of the important developments and additional research needs in censored survival analysis for the reader who is interested in the field but not fully familiar with it. Accordingly, I have attempted to make the level of presentation simple and, as often as possible, descriptive rather than mathematical.

It should be noted that this paper is not intended to be a review of the entire statistical literature on censored survival data. Indeed, I will restrict attention to problems involving general right censoring, and only those where no distinction is made between possibly different causes for the censoring. It is also assumed that censoring for any one subject does not occur for reasons related to the outcomes of other subjects, such as in Type II reliability models (*cf* David 1970). Furthermore, important topics within this restricted area such as sequential methods and the analysis of grouped data are not considered. Without loss of generality, it is assumed throughout that all distribution functions which represent time measurements are absolutely continuous with respect to Lebesgue measure.

2. *Basic Formulation*

In the general right-censored survival problem, the life process of a subject can be represented by random variables T, U, and d. The first of these is true survival time *under test conditions.* It is the distribution function, F, of T which we ultimately wish to investigate. T is not in general observed. Instead, we observe the pair (U, d), where U is the observed portion of survival time and d is an indicator of whether U denotes actual survival time ($d = 1$) or a censored record of it ($d = 0$). It follows that $U \leq T$ in general and $U = T$ only when the observation is uncensored.

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The random variable T normally denotes the time elapsed from some time origin to a specific event of interest. For example, T could be the time until death for an advanced cancer patient, measured from his entry into a clinical trial. The random variable U represents the time elapsed between the same origin and the termination of observation, either because the event of interest occurs or because observation must cease, that is be censored, for some reason. A censored observation could arise if a patient were withdrawn from trial or if he were still alive at the time the data were analyzed. For ease of exposition, we will continue to refer to T as survival time or time until failure and to U as observation time, but it is clear that they can represent any random variables which satisfy the inequality $U \leq T$.

A full probabilistic description of the observable pair (U, d) is given by the quantities p, F_D, and F_C, where $p = P(d = 1)$ is the probability that an observation is uncensored, $F_D(u) = P(U \leq u \mid d = 1)$ is the conditional distribution of observation time, given that censoring does not occur, and $F_C(u) = P(U \leq u \mid d = 0)$ is the conditional distribution of observation time when censoring does occur.

Suppose D_u and C_u, respectively, denote the occurrence of an uncensored and a censored observation in a neighborhood of time u. Then

$$P(D_u) = P(U \epsilon N_u, d = 1) = p dF_D(u) \quad \text{and} \quad P(C_u) = P(U \epsilon N_u, d = 0) = (1 - p)\, dF_C(u)$$

where $N_u = (u - du, u)$ and $dF_j(u) = F'_j(u)du$ for $j = D, C$. Given a set of independent realizations of (U, d), it follows that the subsample of failure times is a set of observations from F_D and the censored times form a set of observations from F_C. It is important to realize that the subsample of uncensored times is not a sample from the distribution F. A set of partially censored data only provides direct information about p, F_D, and F_C, the probability elements associated with the observable pair (U, d). The general problem of censored survival analysis can be viewed as relating this information to F.

In order to relate the distribution of survival time to that of the observable pair (U, d), we introduce two additional functions a and B. These are defined by

$$a(u) = p dF_D(u)/dF(u) = P(d = 1 \mid T \epsilon N_u),$$

the conditional probability of an uncensored observation, given T, and

$$dB(u) = (1 - p)\, dF_C(u)/[1 - F(u)] = P(C_u \mid T > u),$$

the conditional probability of a censored observation in N_u, given that true survival time exceeds u. It follows that

$$P(D_u) = a(u)dF(u) \quad \text{and} \quad P(Cu) = [1 - F(u)]\, dB(u).$$

Also, from the condition that a subject will eventually be observed to fail or be censored, we see that F, a, and B are constrained by the equation

$$\int_0^\infty \{a(u)\, dF(u) + [1 - F(u)]\, dB(u)\} = \int_0^\infty [a(u) + B(u)]\, dF(u) = 1.$$

Thus the triplet (F, a, B) specifies both the joint distribution of the observable random variables (U, d) and the distribution of true survival time T.

The likelihood function of (F, a, B) resulting from a set of independent realizations, (u_i, d_i), of (U, d) is proportional to

$$\prod_i \{a(u_i)\, dF(u_i)\}^{d_i} \{[1 - F(u_i)]\, dB(u_i)\}^{1-d_i}.$$

When each subject is exposed to the same censoring experiences but may have a different

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survival time distribution F_i, the likelihood function is

$$\prod_i \{a(u_i)\, dF_i(u_i)\}^{d_i}\{[1 - F_i(u_i)]\, dB(u_i)\}^{1-d_i}.$$

When subjects are exposed to different censoring experiences but have a common survival time distribution, the likelihood function is

$$\prod_i \{a_i(u_i)\, dF(u_i)\}^{d_i}\{[1 - F(u_i)]\, dB_i(u_i)\}^{1-d_i}.$$

This form may be appropriate for subjects treated alike in different experimental settings, such as, for example, in a multi-clinic study of a single drug. It would also be the likelihood function when censoring arises simply because all patients have not failed by the time of analysis and data are analyzed conditionally on the times of patient entry.

3. *Methods Based on Noninformative Censoring*

3.1. *Independent, Nonprognostic, and Noninformative Censoring*

Nearly all statistical methods for censored survival data are based on the assumption that censoring effects are, in a very specific sense, noninformative with respect to the distribution of survival time. A common example of this type of censoring occurs in a clinical trial where survival is measured from entry into the trial and one observes the true survival times of those patients who fail by the time of analysis and censored times for those who do not. Clearly in this case the distribution of entry times, which completely determines the functions a and B, is not related to F. For this and many other experimental situations in which likelihood inference methods are used, it is assumed that the contribution to the likelihood function from an observed failure at time u is $dF(u)$ and from a censored observation is $1 - F(u)$. It is then concluded that the likelihood function based on a set of independent observations from a population homogeneous with respect to the survival-time distribution is proportional to

$$\prod_i \{dF(u_i)^{d_i}[1 - F(u_i)]^{1-d_i}\} = \prod_j dF(t_j) \prod_k [1 - F(s_k)], \tag{1}$$

where the t_j are observed failure times and the s_k are censored observation times. For heterogeneous populations, the analog of (1) is

$$\prod_i \{dF_i(u_i)^{d_i}[1 - F_i(u_i)]^{1-d_i}\}. \tag{2}$$

The likelihood functions in (1) and (2) are then usually further simplified by taking F to be of a particular form depending on a vector of unknown parameters and, possibly, a set of covariate variables (*cf* Feigl and Zelen 1965, Glasser 1967, Zippin and Armitage 1966 and Byar 1977). This same assumption about censoring is also made in the recent work on regression models for survival data in which partial likelihood functions are used (see, for example, Cox 1972, Kalbfleisch and Prentice 1973 and Cox 1975).

In some instances the assumptions used to arrive at likelihood functions such as (1) and (2) are explicitly formulated, but most often they are either stated in rather vague terms or not stated at all. As a result, it is likely that many users of censored survival methods are either unaware that they are making certain assumptions about the censoring mechanism or are unclear about precisely what these assumptions are. Since these methods are commonly

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applied to data sets in which censoring arises in ways more complex than in the preceding example, it is important that the censoring assumptions required by these methods be clearly understood.

One justification for this treatment of censored data is given by Efron (1967) in what will be called an *independent* censoring model. Here the observable pair (U, d) is structurally represented by

$$U = \min(T, Y) \quad \text{and} \quad d = 1 \quad \text{if} \quad T \leq Y \quad \text{or} \quad d = 0 \quad \text{if} \quad T > Y \tag{3}$$

where the random variable Y is independent of T. Furthermore, the distribution function, H, or Y is assumed to be functionally independent of F. This form of censoring is also referred to as random-censorship when Y is nondegenerate and fixed-censorship when it is regarded as a constant (*cf* Breslow 1970). In some applications the relationship in (3) reflects actual conditions and Y can be interpreted meaningfully. For example, in the clinical trial described earlier, Y is the elapsed time between entry into the study and the time of analysis. In other situations there is no justification of (3) that can be evidenced by the absence of an interpretation for Y, but the representation is still assumed for mathematical convenience.

It follows from the statistical independence of T and Y in (3) that the events D_u and C_u have probabilities

$$P(D_u) = [1 - H(u)]\, dF(u) \quad \text{and} \quad P(C_u) = [1 - F(u)]\, dH(u),$$

and the likelihood function from a homogeneous sample is proportional to

$$\left\{ \prod_j dF(t_j) \prod_k [1 - F(s_k)] \right\} \left\{ \prod_j [1 - H(t_j)] \prod_k dH(s_k) \right\}. \tag{4}$$

Then because H is functionally independent of F, the maximum of (4) can be obtained by separately maximizing each of the bracketed terms. Hence for the purpose of obtaining the maximum likelihood estimator (*MLE*) of F, one need only consider (1).

By arguments similar to the foregoing, one can show that (2) is a valid starting point for a likelihood analysis when subjects have different survival time distributions, but for each (3) holds. There is a second justification of (1) and (2) whenever the condition

$$P(T \epsilon N_t \mid C_u) = dF(t)/[1 - F(u)] \tag{5}$$

holds for all $0 < u \leq t$. This states that a censored observation at time u provides only the information that true survival time exceeds u. In this sense the event of censoring carries no prognostic information about subsequent survival time, and so it is reasonable to refer to any model satisfying (5) as a *nonprognostic* censoring model. It has been shown that independent censoring models form a proper subset of nonprognostic censoring models but, nevertheless, that (1) and (2) are still appropriate for likelihood inference based on any model in the latter class (Williams and Lagakos 1977). Thus models for nonprognostic censoring are more general than ones for independent censoring, and do not require the interpretation of a variable Y representing potential censoring time.

A third condition which leads to (1) and (2) is the constant-sum property, defined by Williams and Lagakos (1977) as

$$a(u) + B(u) = 1 \text{ almost everywhere } F. \tag{6}$$

They have showed that constant-sum censoring models include nonprognostic censoring models (and thus independent censoring models) as a proper subset and that (6) uniquely characterizes all situations for which (1) or (2) is a valid starting point for the likelihood analysis of F. Models satisfying (6) will therefore be called *noninformative* censoring models

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since they allow the functional forms of a and B to be disregarded in likelihood inference for F. Note that when (6) holds, B has the properties of a distribution function.

Each of these three models for censored survival data can be used to justify the commonly used methods of censored-survival analysis, yet each has its own unique value. Independent censoring models are an outgrowth of Type I reliability models and accurately describe the censoring in the clinical trial example used at the beginning of this section. Because of its nice mathematical properties, the independence formulation is quite useful in asymptotic and other theoretical investigations of statistical methods based on noninformative censoring. The primary value of the nonprognostic censoring condition (5) lies in the fact that it does not require the consideration of potential censoring times, yet has a straightforward interpretation. There are a considerable number of causes for censoring which are not representable by the formulation in (4), but may be such that an experienced analyst can relate to and be comfortable with the nonprognostic condition given in (5). Noninformative censoring, as characterized by (6), is the least easily interpreted condition of the three in terms of the physical process, yet it is the most general one mathematically. Accordingly, its primary role is likely to be in those situations where one cannot confidently indentify the censoring mechanism as being independent or nonprognostic, and hence must resort to statistical tests to assess the effects of the censoring mechanism. This point is discussed in more detail in Section 4.

3.2. Estimation of F

It is not surprising that nearly all analyses of data subject to noninformative censoring are based on likelihood methods. Other procedures such as least squares do not seem to be easily adapted to censored data because the outcome of interest is not always observed. Methods based on likelihood functions, however, adapt to censored data quite easily and, as will be shown, can be used to obtain both estimates and approximate tests of significance for F without direct consideration of the censoring function B $(= 1 - a)$.

We have seen that for purposes of ML estimation of F, attention can be restricted to a reduced likelihood function that depends on F, but not on the censoring function B. When F is fully parameterized, the reduced likelihoods in (1) and (2) become functions of real variables and can be maximized by standard methods (*cf* Zippin and Armitage 1966, Byar 1977). Alternatively, Kaplan and Meier (1958) produced their well-known product-limit estimator by maximizing (1) nonparametrically—that is, over the set of all distribution functions. More recently, and stemming largely from a paper by Cox (1972), there has been a great deal of interest in familes for F which involve both a parameter β of direct interest as well as a function of secondary interest. Here the approach has been to first produce a "partial" likelihood function that depends only on β, and then to make inferences about β by treating the partial likelihood as if it were a true likelihood function (see Cox 1972, Kalbfleisch and Prentice 1973, and Breslow 1974, 1975; also see Cox 1975 and Efron 1977, who show that Cox's partial likelihood function can be expressed as a component of (2)).

Although MLE's for F can be obtained without consideration of the censoring function B, it is important to realize that the distribution of these estimators will, in general, depend on both F and B, and usually in a quite complex ways. Only in the simplest cases have the exact distributions of MLE's been derived, and these are usually so complicated that they are of little practical or theoretical use (see, for example, Bauer 1976 who has considered a model based on exponential survival time and noninformative heterogeneous censoring which can be evaluated numerically only with great difficulty). For certain other models, the asymptotic distributions of the MLE's, which also generally depend on both F and B, have been derived.

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For example, Efron (1967) and Breslow and Crowley (1974) derived asymptotic properties of the product-limit estimator of F. In most situations, however, even the asymptotic distribution of the *MLE* cannot be easily obtained.

Fortunately, it is still possible to estimate the variance of the *MLE* and carry out approximate tests of significance for F, and to do so in a way which does not require the direct consideration of B. To see this, suppose F and B are functionally independent and let the family of possible F be indexed by the vector parameter $\boldsymbol{\Theta}$. It follows that the matrix of negative second derivatives of the overall likelihood function for (F, B) is block diagonal, where the block corresponding to F is simply the negative second derivative matrix, $\mathbf{D}(\boldsymbol{\Theta})$, of the reduced likelihood function in (1) or (2). The matrix $\mathbf{D}(\boldsymbol{\Theta})$ is a function of $\boldsymbol{\Theta}$ and the data and hence $\hat{\boldsymbol{\Sigma}} = [\mathbf{D}(\hat{\boldsymbol{\Theta}})]^{-1}$ can be computed and used to estimate $\boldsymbol{\Sigma}$, the covariance matrix of $\hat{\boldsymbol{\Theta}}$. The use of $\hat{\boldsymbol{\Sigma}}$ is justified by the fact that, under mild regularity conditions, $\boldsymbol{\Sigma}$ is asymptotically given by $\{E[\mathbf{D}(\boldsymbol{\Theta})]^{-1}$. Thus $\hat{\boldsymbol{\Sigma}}$, unlike the inverse sample information matrix $\tilde{\boldsymbol{\Sigma}}$, is an easily computable estimate of $\boldsymbol{\Sigma}$ that does not require the direct consideration of the censoring function B. Moreover, recent results by Hinkley (1978) suggest that it often should be preferred to $\tilde{\boldsymbol{\Sigma}}$ on theoretical grounds as well.

Approximate significance tests for $\hat{\boldsymbol{\Theta}}$ can be carried out by approximating the distribution of $\boldsymbol{\Theta}$ by the normal $(\boldsymbol{\Theta}, \hat{\boldsymbol{\Sigma}})$. Alternatively, approximate significance tests can be based on differences in the maximized log likelihood functions (*cf* Cox and Hinkley 1974). It is easy to show that any test derived this way from (1) or (2) is exactly the same test that results from consideration of the full likelihood function for F and B. A third method of obtaining approximate significance tests is through the use of the score statistic formed from the partial derivatives of the log likelihood function with respect to the components of $\boldsymbol{\Theta}$ being tested (*cf* Cox and Hinkley 1974). As before attention can be restricted to the reduced likelihood function and, provided that the sample negative second derivative matrix is used in place of the information matrix, explicit consideration of B can be avoided.

Under suitable regularity conditions, these three types of significance tests are asymptotically equivalent and exact. They also usually agree quite closely in uncensored problems involving moderately large data sets. For censored data problems, however, their small sample behavior is relatively unexplored. In a recent simulation study (Peace and Flora 1978) comparing several survival models, tests of regression vectors based on likelihood ratios seem to consistently yield higher significance levels than those based on normal theory. Since this study involved a small number of repetitions, the differences might be partially explained by sampling variation. Nevertheless, these results underscore the need for investigations of the small sample behaviour of these tests.

3.3. Tests for the Comparison of Several Samples

A fundamental problem in many biological and medical investigations is a comparison of the survival-time distributions from two or more samples of censored data. This is a common occurrence, for example, in clinical trials where several treatments are being evaluated. There are often other factors such as age, sex, etc, which may also be related to survival time and, therefore, must be dealt with. However, the discussion in this section is restricted to the simpler situation in which the N_i observations, (U_{ij}, d_{ij}) $j = 1, 2, \ldots, N_i$, forming the *ith* sample are homogeneous with respect to both survival time and censoring. Since censoring is noninformative, the censored-survival model for the observations in the *ith* sample is fully specified by (F_i, B_i). The hypothesis of interest, the equality of survival time distribution functions across samples, is given by $H_0: F_1 = F_2 = \cdots = F_K$. It will be seen that a critical determinant in the performance of many K-sample tests is the condition

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$$B_1 = B_2 = \cdots = B_K. \tag{7}$$

When (7) holds, the samples are said to have *equal censoring patterns.* An example where this occurs is a comparative clinical trial where, throughout the accrual period, patients are randomized to all of the treatment arms and noninformative censoring occurs only because some patients have not yet failed by the time of analysis. It should be emphasized, however, that in many circumstances the patterns of censoring will not be the equal. For example, (7) will not hold if treatment arms are added or dropped during the course of a clinical trial or if censoring occurs for reasons that vary among the treatments. Unless specified otherwise, we make no assumptions about the equality of censoring patterns.

When the survival time distribution functions F_i are assumed to be elements of a specified parametric family, testing problems are greatly simplified. In the comparison of two exponential distributions with rate parameters λ_1 and λ_2, the most commonly used test is based on $R = \hat{\lambda}_1/\hat{\lambda}_2$, where $\hat{\lambda}_i = \sum_j d_{ij}/\sum_j u_{ij} = d_{i.}/u_{i.}$ is the MLE of λ_i. The use of this statistic with censored data originated in Type I and Type II reliability experiments. In the latter, the null distribution of R is exactly $F(2d_{2.}, 2d_{1.})$, the F-distribution with $2d_{2.}$ and $2d_{1.}$ degrees of freedom. When there is general right censoring, the null distribution of R is *not* the F-distribution but depends in a complicated way on the censoring functions (see, for example, Bauer 1976). Nevertheless, the distribution of R is usually approximated by $F(2d_{2.}, 2d_{1.})$ for purposes of testing H_0.

A second commonly used test for H_0 in the exponential case can be obtained by the application of likelihood methods to the model $\lambda_i = \exp\{\alpha + \beta Z\}$, where Z is an indicator of whether an observation is from the first ($Z = 0$) or second ($Z = 1$) sample. The hypothesis of equality is now simply $\beta = 0$, and the corresponding approximate normal (0, 1) statistic for testing $\beta = 0$ based on the MLE of β is $Q = (\ln \hat{\lambda}_1 - \ln \hat{\lambda}_2)/(d_{1.}^{-1} + d_{2.}^{-1})^{1/2}$. This test can be viewed as one which treats $\ln\hat{\lambda}_{i.}$ to be approximately Normal with mean $\ln\lambda_i$ and estimated variance d_i^{-1} (*cf* Zelen 1969). One advantage of Q over R is that it readily generalizes to $K > 2$ samples. Also, the results of Zelen and Dannemiller (1961) suggest that Q may be less adversely affected by Weibull departures from exponentiality, although this was not investigated for general right censoring.

Despite the widespread use of Q and R, the only published papers that shed light on their small-sample behavior for general right censoring are investigations which are primarily designed for other purposes (*cf* Lee, Desu, and Gehan 1975). Since the exponential distribution plays such an important role in survival time methodology, extensive investigations of the small-sample behavior of these tests should be undertaken. These should include varying sample sizes, varying patterns and amounts of censoring, and departures from exponentiality.

The preceding discussion has involved only the exponential distribution, but it is clear that likelihood methods also can be applied to other parametric families of distributions. In general, this will lead to asymptotically efficient test statistics which can be formed and evaluated in a way that does not require the explicit consideration or the equality of the censoring functions B_i.

We now turn attention to nonparametric tests for H_0 and begin by considering the effects of censoring in regard to the formation of test statistics. For convenience, attention is restricted to the two-sample problem ($K = 2$).

Since linear rank statistics play such a central role in testing situations where data are not subject to censoring, it is not surprising that nearly all nonparametric tests for H_0 are based on extensions of these statistics to include general right censoring. An obvious difficulty caused by censoring in this regard is that, given a set of data, the true survival times, T_{ij}, and

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hence the corresponding vector of ranks, are not in general observed. As a result, there is no unique generalization of the commonly-used linear rank statistics to censored data.

Gehan (1965) proposed a generalization of the two-sample Wilcoxon statistic which is based on metricized pairwise comparisons, say $\psi_G(j, \ell)$, of the *jth* observation from the first sample with the ℓth observation from the second sample. As with the ordinary Wilcoxon statistic, $\psi_G(j, \ell)$ is assigned the values one or zero if, on the basis of the observations, it can be determined that t_{1j} is larger than or smaller than $t_{2\ell}$. When this ordering cannot be determined from the observations (that is, when the smaller observation time is censored), ψ_G is assigned the value one-half. The overall test statistic, S_G, is then formed by summing the ψ_G values. Specifically, $S_G = 2\sum_j\sum_\ell\psi_G(j, \ell) - 1$. When there are no censored observations, $(S_G + 1)/2$ reduces to the usual Wilcoxon statistic.

Efron (1967) proposed an alternative generalization of the Wilcoxon statistic which attempts to make better use of the data in those pairwise comparisons where $\psi_G(j, \ell) = 1/2$. He suggested the statistic $S_E = \sum_j\sum_\ell\psi_E(j, \ell)$, where $\psi_E(j, \ell)$ is a nonparametric estimate of the conditional probability that $T_{1j} > T_{2\ell}$, given u_{1j}, d_{1j}, $u_{2\ell}$, and $d_{2\ell}$. The metric ψ_E equals ψ_G whenever the latter is zero or one, but differs when $\psi_G = 1/2$. Latta (1977) showed that S_E is essentially the same as the generalized Wilcoxon statistic derived by Peto and Peto (1972) from likelihood arguments

Mantel and Haenszel (1959) and Mantel (1966) considered a statistic that has origins in methods for analyzing several 2×2 contingency tables. The general form of the statistic is $S_M = 0_1 - E_1$, where $0_1 = d_{1.}$ is the number of uncensored observations from the first sample. E_1 can be loosely interpreted as a conditional expected value of 0_1. Strictly speaking, however, it is the sum of several conditional expectations, each computed over a different reference set (see also Peto and Pike 1973, Crowley and Breslow 1974). The statistic S_M also arises as a score statistic from the partial likelihood function introduced by Cox (1972), and as the Logrank statistic put forth by Peto and Peto (1972). When there are no tied or censored observations, the statistic S_M reduces to the Savage statistic based on exponential order scores. Peto (1972), Peto and Peto (1972), and Prentice (1978) considered other rank-invariant test statistics for H_0 as well as general methods for the development and evaluation of linear rank statistics.

We now consider the use of these statistics in the formation of nonparametric tests for H_0, and begin with permutation tests. Suppose that $B_1 = B_2$, that is, that the samples have equal censoring patterns. Then under H_0, $(F_1, B_1) = (F_2, B_2)$ so that the $N_1 + N_2$ observation pairs $\{(U_{ij}, d_{ij}) : i = 1, 2, j = 1, 2, \ldots, N_i\}$ are independently and identically distributed. Thus, if S is any test statistic, an exact significance test for H_0 can be based on S and its permutation distribution generated by the $(N_1 + N_2)!/(N_1!N_2!)$ partitions of the observation pairs into groups of size N_1 and N_2.

The choice of S is based on the types of departures from H_0 that seem plausible. The use of S_G leads to the permutation test proposed by Gehan (1965). The use of a slight modification of S_E and S_M lead to the generalized Wilcoxon tests and permutation Logrank tests suggested by Peto and Peto (1972).

The preceding development indicates quite clearly that the basic premise underlying these and all permutation tests for H_0; namely, that all $(N_1 + N_2)!/(N_1!N_2!)$ partitions of the observation pairs are equally probable, requires the equality of B_1 and B_2 as well as the equality of F_1 and F_2. When the censoring patterns are unequal, the observation pairs from the first sample do not have the same distribution as those from the second sample, even when H_0 is true. As a result, it is required for the validity of all permutation tests for H_0 that the censoring patterns be equal. In particular, the permutation tests based on S_G, (Gehan), S_E (Peto and Peto), and S_M (Peto and Peto) are not valid if $B_1 \neq B_2$.

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The use of permutation tests when censoring patterns are unequal has been discussed by several authors (see, for example, Breslow 1970, Gehan 1965, and Efron 1967). However, there do not seem to have been any extensive analytic investigations into the degree to which small-sample significance levels are distorted when permutation tests are applied to data with unequal patterns of censoring.

Breslow (1970) generalized S_G to $K > 2$ samples and proposed an approximate chi-square test for H_0 based on the generalized statistic and an estimate of its unconditional (non-permutational) covariance matrix. The estimated covariance matrix does not rely on the equality of B_1 and B_2 and hence the test is valid whether or not the censoring patterns are equal. Moreover, the Breslow test seems to have comparable power to its permutational analog in situations where $B_1 = B_2$. This suggests that unless it is quite certain that censoring patterns are equal, the unconditional (Breslow) test based on S_G should be used instead of the permutation test (Gehan). It should be cautioned, however, that both of these tests are susceptible to bias if censoring patterns are unequal and the data are grouped (*cf* Mantel 1967, Breslow 1970).

Efron (1967) considered the statistic S_E and derived its asymptotic unconditional distribution for the general case where B_1 and B_2 are not necessarily equal. He also showed that the variance of S_E can be unstable for certain F and B. This difficulty has resulted in the test being seldom used.

Mantel (1966), Cox (1972), and Peto and Peto (1972) suggested tests for H_0 based on S_G and an estimate of its conditional covariance matrix. In particular, the so-called Mantel-Haenszel test is of the general form $X^2_{MH} = (\mathbf{0} - \mathbf{E})' \mathbf{V}^-(\mathbf{0} - \mathbf{E})$, where the components of $\mathbf{0} - \mathbf{E}$ are similar to S_M and $\mathbf{V}^-$ is a generalized inverse. It has been shown that the distribution of X^2_{MH} under H_0 is approximately X^2_{K-1}, and that the test is valid regardless of whether or not the patterns of censoring are equal (*cf* Mantel 1966, Peto and Peto 1972, Crowley 1973, and Crowley and Breslow 1974).

An approximation to X^2_{MH} which is commonly used because of its simpler form is $X^2_{MP} = \sum_i (0_i - E_i)^2/E_i$, where 0_i and E_i are the components of $\mathbf{0}$ and $\mathbf{E}$. Peto and Pike (1973) and Crowley and Breslow (1974) proved that $X^2_{MP} \leqq X^2_{MH}$ for general K and hence that X^2_{MP} is conservative. Based on their results, the degree of conservatism appears to be minimal whenever patterns of censoring are equal and there are no tied observations. Crowley and Breslow warn, however, that the conservatism might be severe if there are marked differences in censoring patterns. Because of the widespread use of X^2_{MP}, a more extensive study of its small-sample conservatism, particularly when censoring patterns are unequal and ties are present, would be of great interest.

Several authors have studied the power and efficiency of these tests. Generally speaking, tests based on S_M seem to have asymptotic efficiency close to one within a rather broad class of proportional-hazard families when censoring patterns are equal, but can suffer a drop in efficiency when censoring patterns are different (see, for example, Crowley 1973, Peto 1972, Efron 1977, Oakes 1977 and Cox 1975). The simulation studies of Lee, Desu, and Gehan (1975) supported the feeling that S_M has high small-sample efficiency, but these were very limited in scope and did not consider the case where censoring patterns were unequal. There is still a need for more extensive studies of the behavior of S_M in small samples with varying amounts and patterns of censoring as well as grouped data.

3.4. Goodness-of-Fit Methods

My purpose in this section is to examine ways of checking assumptions about classes of models when data are subject to general right censoring. Since censoring and its effects are

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the theme of this paper, a logical starting place for this discussion is the assumption that censoring is noninformative. However, this topic is postponed until Section 4 because of its close connection with models for informative censoring. Here I will discuss only tests for hypotheses concerning the survival time distribution F when censoring is noninformative.

One problem of considerable importance is testing for goodness-of-fit of data to a class of noninformative models in which F belongs to a specified family $\mathfrak{F}$ of survival time distributions. For example, one may wish to test whether the survival time distribution for a set of censored data is exponential. Informal tests based on visual scans are always possible in these circumstances because F can be estimated by the model and also by nonparametric methods such as the Kaplan-Meier product-limit estimator. Alternatively, informal tests can be made by comparisons of the model and empirical estimates of the conditional distribution functions F_D and F_c (Williams and Lagakos 1978).

A formal approach to this problem is to use a hierarchical class of models in which it is assumed that $\mathfrak{F}$ is a subset of a larger family $\mathfrak{F}^*$. For example, a test for exponential $\mathfrak{F}$ can be obtained by choosing $\mathfrak{F}^*$ to be the family of two-parameter Weibull distributions or of three-parameter log gamma distributions. Once $\mathfrak{F}^*$ has been selected, the likelihood methods described in Section 3.2 can be applied in a straight-forward way to test whether F is in $\mathfrak{F}$ or $\mathfrak{F}^* - \mathfrak{F}$. When appropriate regularity conditions hold, tests obtained in this way enjoy all the asymptotic optimality properties of ML procedures and can be computed and evaluated without explicit consideration of the censoring function B.

Hierarchical models can also be used to form tests that are sensitive to weakly structured alternatives. For example, a multi-directional test for exponentiality can be formed by taking $\mathfrak{F}^*$ to be the family of distributions with hazard functions $h(t)$ satisfying

$$h(t) = \lambda_j \text{ and } v_{j-1} \leqq t < v_j,$$

where λ_j is an unknown parameter and $0 = v_0 < v_1 < \ldots < v_K < v_{K+1} = \infty$ and K are specified. The likelihood ratio statistic for the test is easily determined to be

$$2 \sum f_j \ln (f_j/w_j) - 2(\sum f_j) \ln (\sum f_j/\sum w_j),$$

where f_j and w_j are the number of failures and total observed time in the interval $I_j = [v_{j-1}, v_j)$, and all summations are from one to $K + 1$. Thus, approximate significance levels for H can be obtained by comparing the value of the statistic with the percentage points of the K *d.f.* chi-square distribution. Since any continuous alternative to the exponential can be closely approximated by an appropriate choice of K and the I_j, this test is, in theory, very flexible. Practical guidelines for the choice of K and the I_j, however, are not obvious and should be investigated. For a related approach, applicable to grouped data, see Turnbull and Weiss (1978).

In situations where censoring never occurs, an alternative to the use of hierarchical classes of models is the use of goodness-of-fit tests based on statistics which measure a difference between an estimate of F from the model and a nonparametric estimate which does not depend on any assumptions about F. Typical examples are Kolmogorov-Smirnov and Cramér-von Mises type tests (*cf* Kendall and Stuart 1967). Although these types of statistics can also be easily formed when noninformative censoring is present, their exact and asymptotic distributions will usually depend, in a complicated way, on the censoring functions B. As a result, there has been very little done on the adaptation of these tests to general right censoring. The only in-depth work has been published by Koziol and Green (1976). They considered a Cramér-von Mises type statistic based on the Kaplan-Meier product-limit estimator of F and derived its asymptotic distribution for the special case where $B(u) = 1 - [1 - F(u)]^\beta$ with nonnegative β. However, even with this restrictive assumption on the censor-

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ing function, the percentage points of the null asymptotic distribution of the test statistic prove difficult to compute and depend on β.

A second problem of considerable importance in survival analysis is that of testing a set of censored data for goodness-of-fit to a model in which parameters depend on observable covariates. For a general discussion of this problem in relation to the widely used socalled Cox regression model (1972), see Kay (1977). Here I will concentrate only on some specific difficulties caused by censoring.

In principle, one could again resort to the use of hierarchical classes of models in conjunction with likelihood methods to produce tests that avoid the direct consideration of B. To do so, however, would require the specification of possible alternatives to the regression model, and these are often difficult to formulate in terms of $\mathfrak{F}^*$.

A less formal but potentially useful approach to the problem is by the consideration of residuals. Nelson (1973) gave a detailed and valuable account of the use of residuals for models in which T, or a transformation of it, can be expressed as the sum of a mean and error term. He suggested residuals be computed as the difference between observed and predicted lifetimes. This yields a set of residuals, some of which are right-censored, whose distribution approximates that of a censored sample from the error distribution in the original model.

A second definition arises from the notion of generalized residuals introduced by Cox and Snell (1968) (see, for example, Crowley and Hu 1977, Kay 1977). Let the survival time distribution function for the *ith* subject in a sample be denoted by $F(t|Z_i)$ to indicate its functional dependence on a regressor vector Z_i. Also, let T_i be the corresponding survival time random variable and $\hat{F}$ be the MLE of F. Then since $-\ln [1 - F(T_i|Z_i)]$ has the unit exponential distribution, it can easily be shown that $\{-\ln [1 - F(U_i|Z_i)], d_i\}$ is distributed as an observation pair from a noninformative censoring model with unit exponential survival time. Hence the transformed data (u_i^*, d_i), where $u_i^* = -\ln [1 - \hat{F}(u_i|Z_i)]$, can be regarded as a set of residuals. When the regression model is correct, they should resemble a censored sample from the unit Exponential. An advantage of this definition of residuals is that their approximated distribution under the correct model is always that of an i.i.d. censored sample from a specified distribution.

Residuals derived from censored-survival data are usually used in two ways. The first consists of comparing a nonparametric estimate of their distribution function with their approximate theoretical distribution when the model is correct. For example, the product-limit estimate of the distribution function of generalized residuals should resemble the function $1 - \exp(-t)$ when the model is correct.

A second treatment of residuals is to plot and examine their individual values. One method is simply to plot the raw residuals, using symbols to distinguish the censored and uncensored values. Alternatively, the censored residuals can be adjusted to approximate their unknown complete values. Crowley and Hu (1974), in an analysis using generalized residuals, added to each censored value ln 2, the expected median additional lifetime. This seemed to produce a better fit than using the expected mean additional lifetime (=1). Although both raw and adjusted residual plots of this type can be useful when there are small amounts of censoring, their value diminishes as the extent of censoring increases.

An alternative approach to the construction of residuals from a set of censored-survival data is to condition on the type (censored or uncensored) of outcome and form *uncensored residuals* with respect to the distribution functions F_D and F_C. To see this more clearly, consider a sample which is homogeneous with respect to the censoring function B, but where F may depend functionally on a regressor vector Z—denoted $F(t|Z)$. The quantities p, F_D, and F_C will also depend on Z and are given by

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$$p_Z = \int_0^\infty [1 - B(u)]\, dF(u|Z),\ F_D(t|Z) = p_Z^{-1} \int_0^t [1 - B(u)]\, dF(u|Z),$$

$$\text{and } F_C(t|Z) = (1 - p_Z)^{-1} \int_0^t [1 - F(u|Z)]\, dB(u).$$

It follows that the conditional distribution of $-\ln [1 - F_D(U|Z)]$ given $d = 1$, and the conditional distribution of $-\ln [1 - F_C(U|Z)]$ given $d = 0$, are each unit exponential. In order to use these results to create residuals, suppose $\hat{F}(t|Z)$ denotes the MLE of $F(t|Z)$. If B is known, as may be the case in the example at the start of Section 2, it need not be estimated. If it is unknown, it can be estimated, for example, by the produce-limit method by interchanging the roles of censored and uncensored observations. Then if Z_i denotes the value of the regressor vector for the observation (u_i, d_i), the observation time can be transformed by $-\ln [1 - \hat{F}_D(u_i|Z_i)]$ if $d_i = 1$ and by $-\ln [1 - \hat{F}_C(u_i|Z_i)]$ if $d_i = 0$, where $\hat{F}_D$ and $\hat{F}_C$ are obtained by substituting $\hat{F}$ and, if need be, $\hat{B}$ for their theoretical counterparts in the formulae for F_D and F_C. This will yield two sets of residuals, one from the uncensored and the other from the censored observations. When the regression model is correct, both should resemble uncensored samples from the unit exponential distribution. Thus an alternative to the use of censored residuals—which directly reflect the fit or lack-of-fit to $F(t|Z)$—is the use of uncensored residuals which reflect $F(t|Z)$ through the functions F_D and F_C. In theory, the two sets of uncensored residuals can be used in much the same way as residuals from data not subject to censoring. Their practical usefulness, however, has not been explored and remains an open question.

4. *Informative Censoring*

In many experimental situations, censoring can be clearly recognized to be noninformative and, as a result, standard methods of analysis can be employed without reservation. In other situations, however, it is not as apparent that censoring is noninformative, and in fact it is sometimes quite clear that the act of censoring is related to ultimate survival time. Three examples of situations I have encountered where the use of noninformative censoring methods is questionable are:

(1) a clinical trial in which some patients remove themselves from study for reasons possibly related to therapy and thereby censor their survival time under test conditions;

(2) a clinical trial in which those patients experiencing a specific critical event such as metastatic spread of disease are, by design, removed from study and no longer followed for survival time;

(3) a clinical trial or animal experiment in which failure times from causes of secondary interest are recorded as censored observations of the failure time from the cause of primary interest.

The first of these examples represents so-called study dropouts. These occur, in most studies, in frequencies which depend on the severity of the therapies being tested and on the degree to which the study protocol is enforced. The second example typifies a common occurrence in clinical trials that were conducted several years ago, but now occurs much less frequently. The third example represents a problem of competing risks (David and Moeschberger 1978) in which it is desired to estimate the distribution of time until failure from a particular cause when other causes of failure are not in effect. This type of censoring is qualitatively different from the others in the sense that complete observation of the survival time of interest is, from an experimental point of view, an unachievable event. This point is considered again in Section 4.4

It is clear that these examples represent situations where the reasons for censoring might

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be related to ultimate survival time, and so the use only of standard methods of analysis based on noninformative censoring would be inappropriate. Accordingly, it becomes important to address the following questions: (*i*) Can a set of censored survival data be tested to determine whether the censoring is noninformative, (*ii*) what could be the consequences of falsely assuming that censoring is noninformative in the estimation of F, and (*iii*) what alternative kinds of analyses are available in these circumstances?

4.1. Tests for Noninformative Censoring

Without making any assumptions about the functions F, a, and B, any attempts to test for the nature of the censoring mechanism must fail because of problems of nonidentifiability associated with the censored survival model (F, a, B). The results of Cox (1959) and Tsiatis (1975) show that there always exists independent censoring models consistent with any probability distribution for the observable pair (U, d). In a similar vein, Williams and Lagakos (1977) proved that the true censored survival model (F, a, B) is always a member in an infinite set of observably indistinguishable models, and that exactly one member of this set is a noninformative (i.e., constant-sum) model. Hence it is not in general possible to construct a statistical test for the alternatives of non-informative and informative censoring.

The only means of overcoming problems of nonidentifiability is to restrict the family, $\mathfrak{F}$, of models (F, a, B) under consideration. For example, Williams and Lagakos (1977) showed that for exponential survival time, censoring is noninformative if and only if (*i*) $E(U) = pE(T)$ and (*ii*)

$$F_D'(u) = [1 - F_U'(u)]/E(U),\ 0 \leq u < \infty \text{ where } F_U(u) = P[U \leq u].$$

The first of these conditions cannot be tested from a set of data, but the second always can since it depends only on functions of the observable pair (U, d). This means that alternatives to noninformative censoring characterized by (*ii*) cannot be detected but that all other alternatives can be detected.

In order for problems of nonidentifiability with respect to the noninformative property to be totally overcome, it is necessary and sufficient to restrict $\mathfrak{F}$ so that every informative model in the restricted family gives rise to a distribution for (U, d) which is different from that generated by the noninformative censoring model. That is, if (F^*, a^*, B^*) denotes the noninformative model in $\mathfrak{F}$, there can be no other $(F, a, B) \epsilon \mathfrak{F}$ satisfying $a(u)\, dF(u) = a^*(u)\, dF^*(u)$ and $[1 - F(u)]\, dB(u) = [1 - F^*(u)]\, dB^*(u)$ for every u. This point is considered again in Section 4.3.

4.2. Consequences of Falsely Assuming a Noninformative Model

Since almost all methods of analysis for censored-survival data rely on the assumption that censoring is noninformative, it is important to be aware of the possible consequences of making this assumption when it is false. Fisher and Kanarek (1974) considered a censored survival model in which the act of censoring coincides with an event which alters subsequent survival time by a scale parameter α. For $\alpha = 1$ the model corresponds to noninformative censoring, while for $\alpha > 1$ ($\alpha < 1$), it corresponds to a situation in which censoring carries a favorable (unfavorable) prognosis for subsequent survival. Furthermore for each specified value of α, a nonparametric estimate of F can be obtained from a set of censored data. Thus, one can estimate F for each of several choices for α and determine the possible effects of informative censoring.

Peterson (1977) generalized the results of Fisher and Kanarek by deriving general bounds

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for F which do not depend on any assumptions about the type of censoring. Moreover, the bounds can be estimated from a set of censored data and therefore used to assess the possible consequences of falsely assuming noninformative censoring.

Williams and Lagakos (1977) show that when survival time is Exponential with rate parameter λ, the family of nonidentifiable F is indexed by lambda in an interval $[\lambda_1, \lambda_2]$ whose endpoints are estimable from a set of censored survival data. Within this family the element $\lambda = \lambda_1$, corresponds to noninformative censoring while larger λ correspond to models with increasing degrees of informative censoring. The right-hand endpoint represents the limiting case where censoring, when it occurs, immediately precedes failure. These results indicate that the consequences of assuming noninformative censoring when survival is Exponential are to underestimate λ and hence to overestimate median survival time. Moreover, the percentage underestimate of λ is bounded by $100(1 - p)$. Thus, for example, if only 10% of the observations are censored, an unbiased estimator of λ based on noninformative censoring will be in error, on the average, by at most 10%.

4.3. Informative Censoring Models

There have been two approaches to the formulation of classes of censored survival models for situations where there is no strong reason to believe that censoring is noninformative. The first consists of a recourse to a representation like the one in (3) but with a bivariate distribution for (T, Y) for which independence of T and Y is a special case. Provided each point in the parameter space of this joint distribution corresponds to a distinct distribution for (U, d), all problems of nonidentifiability are avoided. This allows the ML estimation of the parameters of the joint distribution and, from these, tests for noninformative censoring as well as estimates for F which properly account for censoring. Moeschberger (1974) suggested this approach in a competing risks setting and considered both the bivariate normal and a bivariate Weibull distribution for (T, Y). Gail (1974) adopted a similar approach, also in the context of competing risks.

Lagakos and Williams (1978) approached the problem by proposing a model for the censoring functions a and B in terms of an unspecified relative-odds-for-failure function c and a scalar θ which reflects the degree to which censoring affects survival. The parameter θ spans the unit interval and signifies noninformative censoring when $\theta = 1$ and informative censoring when $0 \leq \theta < 1$. The limiting case $\theta = 0$ corresponds to censoring which is immediately followed by failure. Conditions for elements in a class of these models to be observably distinguishable have a particularly simple representation in terms of the functions c and F. When survival time is exponential, the rate parameter is identifiable whenever c is nonconstant, a condition easily assessed on the basis of a set of data. The authors considered exponential survival in detail and used likelihood methods to assess the data for fit to a noninformative censoring model and to estimate the failure rate parameter.

In both of these approaches, the price paid for eliminating problems of parameter nonidentifiability is the introduction of untestable assumptions, such as a bivariate Weibull distribution for (T, Y) or exponential distribution for T. As a result, the robustness of these methods to departures from these assumptions is an important consideration and, to my knowledge, has not been examined.

4.4. Discussion

It is worthwhile to mention some practical considerations associated with the analysis of survival data when censoring is informative. Situations similar to the second example given

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at the start of this section should be prevented whenever possible. This can be done by avoiding study designs of this type altogether, or by basing treatment evaluations on endpoints other than survival time. When neither of these alternatives is possible, every effort should be made to obtain survival time follow-up information for all patients removed from the study. This data can then be combined with that of patients not removed from study to estimate the distribution of survival time. Estimates obtained in this way will tend to give distorted impressions of F, the distribution of survival time under test conditions. Nevertheless, they may still be beneficial and can be used in conjunction with estimates of F based on informative censoring models for an overall appraisal of survival time.

It has been my experience that, despite any and all preventive measures, censored observations of the type in the first example at the start of this section are often unavoidable. When the amount of censoring is small, very little bias is likely to result from the use of methods based on noninformative censoring. However, when informative censoring is extensive, substantial biases can occur, and so informative censoring models should be considered. In either case, the extent of possible bias by use of an incorrect censoring model can always be assessed by the methods in Section 4.2.

Censored observations of the type described in the third example cannot usually be prevented since they represent failures from secondary causes. As mentioned earlier, this type of censoring is different from most others because the hypothetical failure time being censored often cannot be verified in any experimental sense. Because of this ambiguity, a statistician faced with this type of data should first determine whether an inference about F is actually necessary. Often an analysis based only on the distribution of (U, d) is sufficient. This might be the case, for example, in a study of time and patterns (local versus metastatic) of disease recurrence in a population of cancer patients. In other situations, however, it is useful to also view competing risks data as a censored data problem. For example, it may be of interest to consider the possible effects on time to metastatic failure from an effective radiotherapy treatment which eliminates local failures. Here the exclusive use of methods based on noninformative censoring can lead to misinterpretations of the data and models based on informative censoring should be considered as part of an arsenal of analytic techniques.

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